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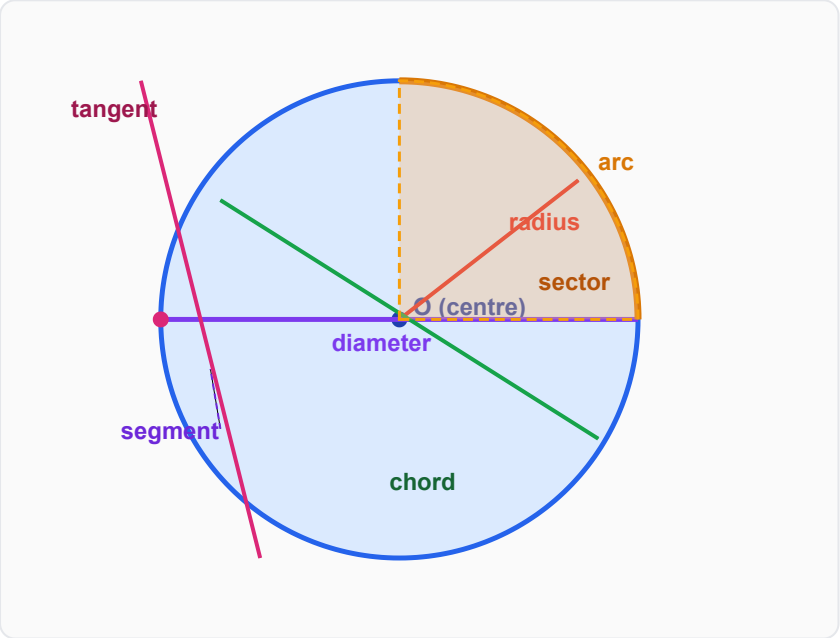
4.5 Circles, Circle Theorems & Constructions

OCR GCSE Mathematics J560 — Component 4 (Geometry and Measures) — Papers 1, 2, 3

4.5.1 Parts of a Circle



Part	Definition
Radius	Line from centre to circumference
Diameter	Chord passing through the centre; equals 2 × radius
Chord	A straight line joining two points on the circumference
Arc	Part of the circumference between two points
Sector	Region bounded by two radii and an arc ("pizza slice")

Segment	Region bounded by a chord and an arc
Tangent	A straight line that touches the circle at exactly one point

4.5.2 Arc Length and Sector Area

For a sector with radius r and angle θ (in degrees):

$$\text{Arc length} = \frac{\theta}{360} \times 2\pi r$$

$$\text{Sector area} = \frac{\theta}{360} \times \pi r^2$$

Memory aid: Both are just a fraction of the full circle ($\frac{\theta}{360}$) multiplied by the full circumference or full area.

Worked Example: A sector has radius 9 cm and angle 80°. Find (a) arc length and (b) sector area. Give exact answers in terms of π.

(a) Arc $= \frac{80}{360} \times 2\pi \times 9 = \frac{2}{9} \times 18\pi = \mathbf{4\pi}$ **cm** ≈ 12.6 cm

(b) Area $= \frac{80}{360} \times \pi \times 9^2 = \frac{2}{9} \times 81\pi = \mathbf{18\pi}$ **cm²** ≈ 56.5 cm²

Worked Example: **Reverse problem:** A sector has area 50 cm² and radius 7 cm. Find the angle.

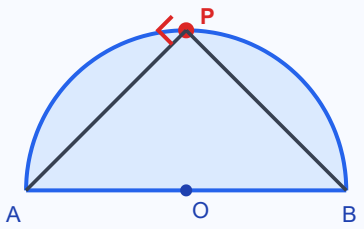
$$50 = \frac{\theta}{360} \times \pi \times 49$$
$$\frac{\theta}{360} = \frac{50}{49\pi} = 0.3249...$$
$$\theta = 0.3249 \times 360 \approx \mathbf{117^\circ}$$

Exam Tip: "Leave in terms of π" means do not press = on your calculator for the π step. Give the answer as e.g. 18π cm². The perimeter of a sector = arc length + 2 radii (many students forget the two radii).

4.5.3 Circle Theorems

You must know, apply, and **state by name** all seven circle theorems.

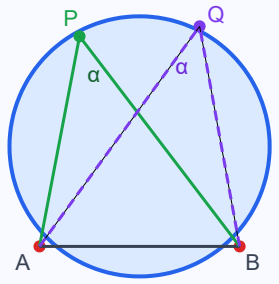
Theorem 1: Angle in a Semicircle



Theorem: The angle in a semicircle is always 90°.

If AB is a diameter, then angle APB = 90° for any point P on the circle.

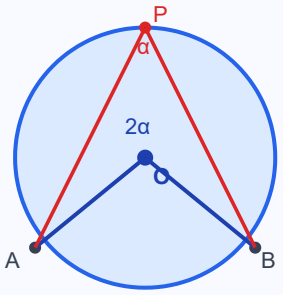
Theorem 2: Angles in the Same Segment



Theorem: Angles subtended by the same arc (or chord) in the same segment are equal.

Angle APB = Angle AQB

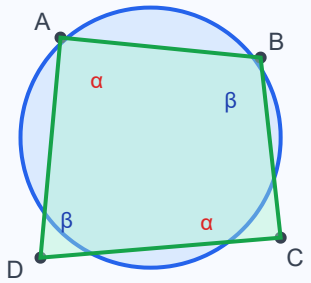
Theorem 3: Angle at the Centre



Theorem: The angle at the centre is twice the angle at the circumference when both are subtended by the same arc.

Angle AOB = 2 × Angle APB

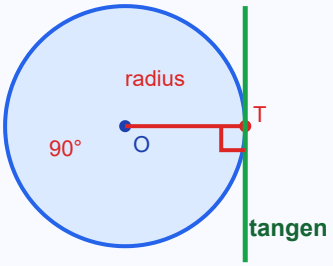
Theorem 4: Cyclic Quadrilateral



Theorem: Opposite angles in a cyclic quadrilateral sum to 180°.

Angle A + Angle C = 180° and Angle B + Angle D = 180°

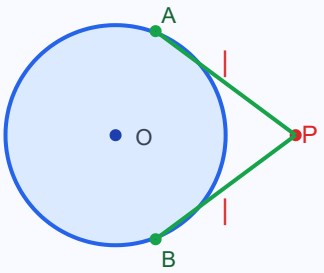
Theorem 5: Tangent–Radius



Theorem: A tangent to a circle is perpendicular to the radius at the point of tangency.

Angle OT (tangent) = 90°

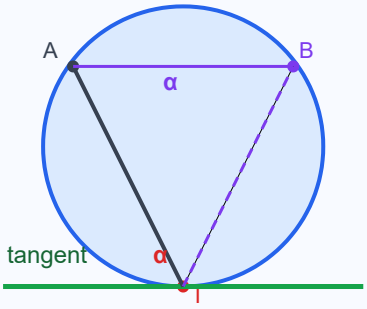
Theorem 6: Tangents from an External Point



Theorem: Tangents from an external point to a circle are equal in length.

PA = PB

Theorem 7: Alternate Segment Theorem (Tangent–Chord Angle)



Theorem: The angle between a tangent to a circle and a chord drawn from the point of tangency equals the inscribed angle on the opposite side of the chord.

Angle between tangent and chord TA = Angle ABT (in alternate segment)

This theorem is commonly tested and requires careful identification of which angle is in the "alternate" segment.

Exam Tip: You must state the theorem by name to gain the reasoning mark. For example: "Angles in the same segment are equal" or "Opposite angles of a cyclic quadrilateral sum to 180°." One mark is always awarded for correct reasoning.

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Worked Example: O is the centre. Angle AOB = 130°. Find angle ACB (C is on the major arc).

The angle at the centre (130°) is twice the angle at the circumference in the same segment.
But C is on the major arc, so angle ACB is subtended by the minor arc.
Reflex angle AOB = 360° − 130° = 230°
Angle ACB = 230° ÷ 2 = **115°**
(Angle at centre = 2 × angle at circumference, applied to the major arc.)

4.5.4 Constructions

All constructions must be done with **compass and straightedge only**. Leave construction arcs visible — do not erase them.

Construction	Steps
Perpendicular Bisector of Line AB	1. Open compass to more than half of AB. 2. Place at A, draw arcs above and below the line. 3. Same compass width, place at B, draw arcs to intersect previous arcs at X and Y. 4. Join X to Y — this is the perpendicular bisector.
Angle Bisector of Angle BAC	1. Place compass at A, draw an arc cutting AB at P and AC at Q. 2. Place compass at P, draw an arc inside the angle. 3. Same width at Q, draw an arc to intersect the previous arc at R. 4. Join A to R — this bisects the angle.
Perpendicular from Point P to Line	1. Place compass at P, draw two arcs cutting the line at A and B. 2. Construct the perpendicular bisector of AB — this passes through P.

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Equilateral Triangle / 60° Angle	1. Draw line AB. 2. Set compass to AB length. 3. Draw arc from A (centre A, radius AB). 4. Draw arc from B (centre B, radius AB) — arcs intersect at C. 5. Join AC and BC. Triangle ABC is equilateral. Angle CAB = 60°.

Exam Tip: Construction marks are awarded for accuracy (within 2mm and 2°) AND for leaving visible arcs. Rubbing out your arcs will lose marks even if the final line is correct. Always use a sharp pencil.

4.5.5 Loci

A **locus** (plural: **loci**) is the set of all points satisfying a given condition. Loci are usually drawn as lines or regions.

Condition	Locus	How to construct
Fixed distance d from a point P	Circle of radius d centred at P	Compass set to d, centre at P
Fixed distance d from a line segment AB	Rectangle with semicircles at each end ("stadium" shape)	Parallel lines distance d from AB, semicircles of radius d at A and B
Equidistant from two points A and B	Perpendicular bisector of AB	Perpendicular bisector construction
Equidistant from two lines	Angle bisector of the angle between the lines	Angle bisector construction

Worked Example: A goat is tethered to a post P by a 4 m rope. It can also reach a wall that is 2 m from P. Shade the region the goat can graze.

The locus of points within 4 m of P is the circle of radius 4 m centred at P. The goat cannot enter the wall, so the region is the arc of the circle on the side away from the wall, cropped by the wall boundary.

Exam Tip: Read loci questions carefully: "within 3 m" means shade the region inside the circle, not just the boundary. "Closer to A than to B" means shade the side of the perpendicular bisector that contains A.

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